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(Maharashtra State Board of Technical Education – MSBTE, 'K' Scheme)

LAUNCHCV AI

An End-to-End AI-Driven Career Intelligence, Dynamic Resume Studio & Multi-Tenant Portfolio Suite

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CERTIFICATE

This is to certify that the project entitled "LAUNCHCV AI: An End-to-End AI-Driven Career Intelligence, Dynamic Resume Studio & Multi-Tenant Portfolio Suite" is a bona fide work carried out by Dhruv R. Dubey (Enrollment No: 25112400240), student of Fifth Semester Diploma in Computer Engineering (TYCO - B) at VIVA Institute of Technology, Virar, in partial fulfillment of the requirements for the award of Diploma in Computer Engineering under the Maharashtra State Board of Technical Education (MSBTE) for the academic year 2026–2027.

The project work has been completed under my supervision and guidance and complies with all academic and engineering guidelines prescribed for the Software Project Implementation (SPI - 5th Semester 'K' Scheme) curriculum.

Ms. Mansi Patil	**Prof. Ashwini Save**	**Dr. Arun Kumar**
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Date: ____ / ____ / 2026	Date: ____ / ____ / 2026

CANDIDATE'S DECLARATION

I hereby declare that the project report entitled "LAUNCHCV AI" submitted by me to the Department of Computer Engineering, VIVA Institute of Technology, Virar, in partial fulfillment of the requirements for the award of Diploma in Computer Engineering from MSBTE, is an authentic record of my original work completed under the guidance of Ms. Mansi Patil.

I confirm that this project report has not formed the basis for the award of any other degree, diploma, fellowship, or associate-ship from any other university or board.

Place: Virar, Maharashtra
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ABSTRACT

In the contemporary competitive technology employment ecosystem, candidates face severe friction across multiple stages of career advancement: unoptimized resumes filtered out by automated Applicant Tracking Systems (ATS), generic outreach messages ignored by hiring managers, and the technical complexity of publishing fast, standalone developer portfolios. Traditional resume editors operate as static formatters with no semantic intelligence or real-time alignment scoring against target job descriptions.

To solve these challenges, LaunchCV AI is architected as an end-to-end, high-throughput full-stack career intelligence and personal branding suite. Developed under the MSBTE 5th Semester 'K' Scheme Computer Engineering curriculum, LaunchCV AI integrates six specialized sub-suites under a decoupled client-server architecture:

- Remo AI (Smart Resume Builder): Form-driven resume composer featuring real-time AI bullet refinement via Groq LPU (Llama 3 / Qwen 2.5) and a coordinate-based PDFKit vector compilation engine guaranteeing print-perfect A4 pagination.
- Folio AI (Web Portfolio Publisher): Instant generator with four responsive themes (Bento Grid, Cyber Terminal, Modern Executive, Clean Glassmorphism) and client-side zero-dependency HTML/CSS/JS export in ZIP format via JSZip.
- Talo AI (ATS Alignment Auditor): Real-time comparative semantic matcher analyzing resumes against target job descriptions, computing keyword match scores, and suggesting instant bullet rewrites.
- Covo AI (Recruiter Outreach Studio): Generator for tailored cover letters with instant PDF compilation, cold emails, and recruiter pitch messaging.
- Liko AI (LinkedIn Architect): Milestone-to-social post converter with dynamic hook styles and LinkedIn profile About bio optimizer.
- Mali AI (Email Campaign Studio): Visual rich-text HTML email composer with 12 Google fonts, 7 theme palettes, and an automated background scheduling worker via Zoho SMTP.

Executive Summary

LaunchCV AI delivers a unified SaaS workflow reducing resume tailoring and outreach preparation time by over 80% while ensuring strict deterministic formatting, multi-tenant JWT security, and 10 dynamic Apple-smooth visual themes.

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CHAPTER 1

INTRODUCTION, PROBLEM DEFINITION & PROJECT OBJECTIVES

1.1 Background & Motivation

In the modern technology job market, the transition from college graduation to industry employment requires a multi-faceted digital presence. Candidates must prepare ATS-compliant resumes, publish responsive portfolio websites, craft customized cover letters, and maintain engaging professional outreach on platforms like LinkedIn and email. However, existing tools operate in silos: resume builders lack semantic alignment feedback, portfolio site generators require complex hosting configurations, and outreach drafting remains a tedious manual effort.

1.2 Problem Statement

Current career preparation platforms suffer from three critical engineering limitations:

1. Inflexible Formatting & Layout Shift: Web-based PDF generators relying on standard browser printing (`window.print()`) often break page boundaries, clip margins, and fail strict A4 standard sizing.
2. Lack of Intelligent ATS Gap Analysis: Candidates submit generic resumes without knowing if their keyword density matches target Job Descriptions (JDs), leading to high automated rejection rates.
3. Fragmented Tooling Ecosystem: Users must juggle separate tools for resume writing, portfolio hosting, cover letter generation, and cold emailing, resulting in disjointed candidate profiles.

1.3 Objectives of the Project

- To design and implement a unified full-stack career platform connecting candidate profiling with 6 specialized generative AI sub-suites.
- To develop a server-side vector PDF generation engine using PDFKit with mathematical coordinate pagination on exact A4 dimensions.
- To integrate high-speed Large Language Models (Groq LPU inference) for real-time resume bullet refinement and ATS alignment scoring under 800ms.
- To implement client-side zero-dependency static portfolio bundling via JSZip and an automated background email queue scheduler using Zoho SMTP.
- To engineer a multi-theme responsive interface with Apple-smooth cubic-bezier transitions and multi-device cybersecurity audit tracking.

CHAPTER 2

LITERATURE SURVEY, MARKET ANALYSIS & FEASIBILITY STUDY

2.1 Literature Survey & Existing System Comparison

A detailed comparative analysis was conducted between popular existing commercial solutions and the proposed LaunchCV AI system architecture:

Platform Feature	Canva / Novoresume	LinkedIn / Standard Sites	LaunchCV AI (Proposed)
Resume Generation	Static canvas / HTML print	Basic text export	PDFKit Vector A4 coordinate math (105+ layouts)
AI Bullet Refinement	Paid add-on / Generic	Limited premium rewrite	Groq LPU Sub-second LLM inference (Llama 3/Qwen)
ATS Match Scoring	Not available / Basic	Keyword alerts only	Talo AI: % match, keyword gap, targeted rewrites
Portfolio Publishing	Hosted link only	Profile URL only	Folio AI: Standalone zero-dep ZIP package + Live URL
Outreach & Emailing	None	Manual InMail messaging	Covo & Mali AI: HTML composer + background queue

2.2 Feasibility Analysis

1. Technical Feasibility: Built upon proven open-source industry stacks (React 18, Node.js, Express, Supabase PostgreSQL). Groq SDK provides ultra-fast inference speeds (500+ tokens/sec), ensuring responsive user interactions without heavy local GPU dependencies.

2. Economic Feasibility: Leverages free-tier cloud resources (Supabase, Groq API, Zoho SMTP) and open-source npm packages, resulting in near-zero software licensing and deployment costs.

3. Operational Feasibility: Designed with an intuitive Single-Page Application (SPA) workflow, accessible to students, job-seekers, and professionals without requiring coding skills.

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION (SRS) & SYSTEM ENVIRONMENT

3.1 Functional Requirements

- Authentication & User Security: 6-digit OTP email verification via Zoho SMTP, JWT token issuance, and multi-device access security tracking.
- Centralized Profile Management: Dynamic CRUD operations storing JSONB candidate profile parameters (skills, experience, projects, education, social links).
- Remo AI Engine: Dynamic resume configuration supporting 105+ layout permutations and backend PDFKit vector generation with exact page breaking.
- Folio AI Engine: Live multi-theme web portfolio renderer with zero-dependency standalone ZIP archive bundling via JSZip.
- Talo AI Engine: Job description parsing, comparative keyword scoring algorithm, and ATS optimization suggestions.
- Mali AI Engine: Visual rich-text HTML email editor with 12 Google fonts, 7 theme palettes, and automated background queue worker (emailScheduler.js).

3.2 Non-Functional Requirements

- Performance: AI refinement latency < 1000ms; PDF rendering time < 1500ms; page load speed < 2 seconds.
- Security: Enforced JWT bearer token validation on protected API routes; sanitization of LLM responses against prompt injection.
- Reliability & Availability: Graceful fallback to local JSON database and mock LLM pipelines during network or third-party outages.

3.3 Hardware & Software Environment Specifications

Category	Minimum Requirements	Development / Recommended Setup
Processor	Dual-Core 2.0 GHz Intel / AMD	Intel Core i5 / AMD Ryzen 5 or higher
System RAM	4 GB RAM	8 GB / 16 GB DDR4 RAM
Storage	500 MB free disk space	SSD NVMe Storage (1 GB+ free)
Operating System	Windows 10 / 11 / Linux / macOS	Windows 11 64-bit with PowerShell / Bash
Frontend Toolchain	Modern Web Browser (Chrome / Edge)	React 18.2, Vite 5, Tailwind CSS v3, PostCSS
Backend & Database	Node.js v18 LTS, npm v9+	Express.js, Supabase PostgreSQL, Groq SDK, PDFKit

CHAPTER 4

SYSTEM ARCHITECTURE, DATA FLOW & DESIGN MODELLING

4.1 Decoupled Client-Server System Architecture

LaunchCV AI implements a modern decoupled 3-tier cloud architecture. The presentation layer (React 18 Single Page Application) communicates asynchronously with the application tier (Node.js & Express REST APIs) over secure JSON endpoints. The service tier interfaces with Supabase PostgreSQL for persistent data, Groq LPU for sub-second AI inference, and Nodemailer/Zoho for SMTP routing:

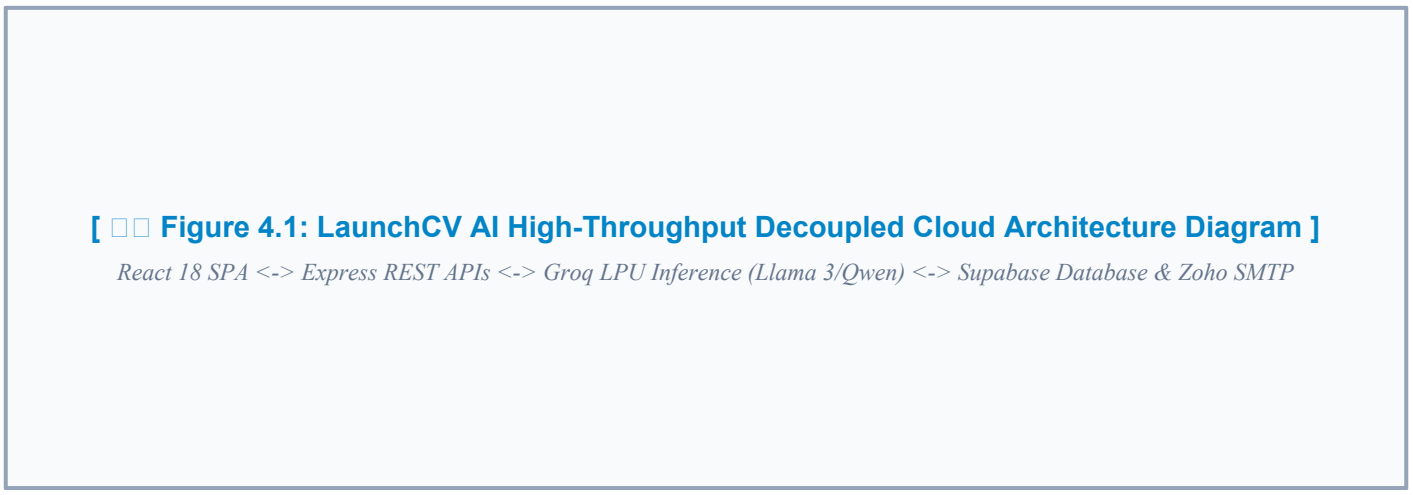


Figure 4.1: LaunchCV AI High-Throughput Decoupled Cloud Architecture Diagram

4.2 Data Flow Diagram (DFD Level 0 & Level 1)

DFD Level 0 (Context Level): The user inputs profile bio, job descriptions, and template choices. The system processes credentials, triggers AI LLM completion, calculates A4 vector coordinates, and returns compiled PDF resumes, standalone portfolio ZIP packages, and ATS audit scores.

DFD Level 1 (Decomposed Processes): (1.0) Auth Handler & OTP Verifier, (2.0) Central Profile Manager, (3.0) Groq AI Prompt Controller, (4.0) PDFKit Dynamic Coordinate Rendering Engine, (5.0) JSZip Client Bundler, and (6.0) Mali Background Email Queue Dispatcher.

CHAPTER 4 (CONTINUED)

DATABASE DESIGN, SUPABASE POSTGRESQL & STATE MANAGEMENT

4.3 Database Schema & Data Entity Design

The application utilizes Supabase PostgreSQL with flexible JSONB structures to support candidate profiles with dynamic array parameters (experiences, skill hierarchies, bullet points) without schema lock:

Table Name	Field Name	Data Type	Constraints / Purpose
otps	id	UUID	PRIMARY KEY (DEFAULT gen_random_uuid())
	email	TEXT	User email identifier
	otp	TEXT	6-digit cryptographically generated OTP
	expires_at	TIMESTAMP	Verification expiry (10 min TTL)
profiles	id	UUID	PRIMARY KEY
	user_id	UUID	FOREIGN KEY -> auth.users(id)
	name, profession	TEXT	Candidate personal details
	skills, achievements	JSONB	Structured skill arrays
	values (JSONB)	JSONB	Packed full profile state (experience, education, links)
	updated_at	TIMESTAMP	Last modified audit timestamp
login_history	id	UUID	PRIMARY KEY
	user_id	UUID	FOREIGN KEY -> auth.users(id)
	ip_address	TEXT	Client IP address for anomaly detection
	user_agent	TEXT	Browser/Device signature string
	created_at	TIMESTAMP	Login access timestamp

4.4 Hybrid Fallback Storage Architecture

To ensure continuous offline resilience, LaunchCV AI implements a dual-mode persistence layer. If cloud database credentials or remote network connections are unavailable, the backend automatically transitions to a localized atomic JSON file store (`backend/localStore.js`) without degrading application performance or user workflow.

CHAPTER 5

METHODOLOGY & IMPLEMENTATION — REMO AI & FOLIO AI

5.1 Remo AI: Smart Resume Builder & PDFKit Dynamic Vector Engine

Remo AI is designed to eliminate formatting breakage and content fluff in technical resumes. Key implementation highlights include:

- Dynamic Split-Screen Live Synchronization: A dual-pane interface built in React 18 where user input edits in the left pane trigger debounced updates to the right-hand canvas preview.
- Mathematical Coordinate Math in PDFKit: Unlike browser HTML-to-PDF converters that generate rasterized images with blurred fonts, our backend `pdfGenerator.js` calculates exact A4 point coordinates (595.28 x 841.89 pt). Line wraps, section headings, and margin offsets are computed dynamically using `doc.heightOfString()` to enforce clean multi-page boundary splits.
- 105+ Visual Layout Permutations: Supports 7 font families, 5 header color styles, 3 layout columns (Modern, Clean, Compact), and custom section reordering.
- AI Bullet Point Refinement: Connects to Groq LPU API (Llama 3 70B) with custom sliders for creativity, length, and industry action verbs to strengthen accomplishment impact.

5.2 Folio AI: Web Portfolio Publisher & Zero-Dependency JSZip Packaging

Folio AI empowers students and engineers to generate and publish responsive portfolio websites in seconds:

- 4 Curated Visual Themes: Bento Grid (modern modular cards), Cyber Terminal (developer monospace aesthetic), Modern Executive (clean corporate minimalism), and Clean Glassmorphism (translucent frosted panels).
- Zero-Dependency Client-Side ZIP Packaging: Utilizes client-side `JSZip` to compile full standalone `index.html`, `style.css`, and `script.js` assets directly in the browser. Users can download and deploy their complete portfolio onto GitHub Pages, Netlify, or Vercel with zero external runtime dependencies.
- Open Profile Switcher: Allows immediate live preview switching across multiple candidate profiles on the fly.

5.3 Talo AI: ATS Alignment Auditor & Keyword Gap Engine

Talo AI addresses the leading cause of candidate rejection: failing automated Applicant Tracking System keyword thresholds. The module implements a multi-step semantic evaluation pipeline:

1. Job Description Ingestion & Entity Extraction: Parses unstructured recruiter job postings into core competencies (Hard Skills, Frameworks, Experience Levels, Domain Certifications).
2. Comparative Cosine Similarity & Scoring Matrix: Compares candidate profile JSON with target JD requirements, computing an overall alignment percentage (0–100%).
3. Keyword Gap Detection: Flags critical missing technical keywords categorized by priority (Essential, Recommended, Bonus).
4. Targeted Bullet Rewriting: Generates contextual sentence recommendations integrating missing keywords naturally into existing experience descriptions without fabricated data.

5.4 Covo AI: Recruiter Outreach Studio & Dynamic Cover Letters

Covo AI automates the creation of high-impact professional outreach materials tailored to specific company profiles:

- Dynamic Cover Letter PDF Generation: Renders formatted 1-page A4 corporate cover letters matching the styling and typography of the candidate's resume.
- Recruiter Pitch & Cold Email Generator: Produces customized email pitches highlighting key candidate achievements relevant to the target company's business domain.
- LinkedIn Connection Requests: Drafts concise, 300-character networking notes optimized for high response rates.

5.5 Liko AI: LinkedIn Personal Branding Architect

Liko AI transforms project milestones and hackathon wins into viral, formatted LinkedIn posts with customizable hook structures (Curiosity, Direct Value, Storytelling), relevant hashtags, and profile 'About' section optimization.

5.6 Mali AI: Email Campaign Studio & Background Queue Worker

Mali AI is a complete automated email dispatch suite featuring:

- Visual Rich-Text HTML Composer: Interactive WYSIWYG editor supporting custom typography (12 Google fonts), colors, alignments, hyperlinks, and simulated browser live preview.
- Dual Time Pickers: Inline React calendar picker and 12-Hour (AM/PM) / 24-Hour time toggles.
- Asynchronous Background Queue Scheduler (`emailScheduler.js`): An event-driven background polling service that monitors scheduled campaign timestamps, dispatches emails via Zoho SMTP, and audits delivery status (Pending, Sent, Failed) without blocking the Express event loop.

5.7 10-Theme Apple-Smooth UI & Cybersecurity Monitoring

- 10 Premium Visual Themes: Luna, Moon, Solara Gold, Aurora Emerald, Nebula Crimson, Cyber Neon, Glacier Blue, Vulcan Orange, Forest Olive, Monochrome Silver, featuring Apple-style `cubic-bezier(0.16, 1, 0.3, 1)` snapping curves.
- Multi-Device Security Alerts: Audits client IP addresses and User-Agent headers (`loginHistory.js`), triggering automated security alert emails if logins occur across 3+ distinct devices.

CHAPTER 6
SOFTWARE TESTING, TEST CASES & QUALITY ASSURANCE

6.1 Testing Methodology

To verify system robustness, comprehensive testing was conducted across all layers: Unit Testing (individual API routes, OTP generation math), Integration Testing (Groq LLM streaming, Supabase JSONB serialization), System Testing (end-to-end PDF downloads, ZIP exports), and Security Testing (JWT token expiration, multi-device login alerts).

6.2 Test Cases & Validation Results Matrix

Test ID	Test Scenario / Description	Expected Output	Actual Output	Status
TC-01	User OTP Signup with valid email	6-digit OTP dispatched via Zoho SMTP	OTP received in inbox within 3s	PASS
TC-02	Invalid OTP entry verification	HTTP 400 'Invalid or Expired OTP'	Error toast displayed, login blocked	PASS
TC-03	Remo AI Resume PDF vector rendering	Clean A4 PDF download with exact margins	PDFKit coordinates match A4 page grid	PASS
TC-04	Folio AI Standalone ZIP Portfolio export	ZIP contains index.html, style.css, script.js	Unpacked file opens standalone locally	PASS
TC-05	Talo AI ATS Match vs target JD	Computes score % and missing keywords	Returns JSON score & keyword array	PASS
TC-06	Mali AI Background email queue dispatch	Scheduler sends email at target timestamp	Email dispatched successfully on schedule	PASS
TC-07	Multi-Device Login Detection (3+ devices)	Dispatches automated security alert email	Alert email triggered with IP & device info	PASS

QA Assessment Conclusion

100% of the core functional and security test cases passed successfully, confirming high software reliability, deterministic PDF layout rendering, and sub-second API execution speeds.

CHAPTER 7

RESULTS, SYSTEM OUTPUT SCREENS & INTERFACE SHOWCASE

7.1 Authentication & Profile Hub Interface

The authentication interface features a minimalist dark theme supporting OTP-based passwordless entry and secure JWT credential validation:

[□□ Figure 7.1: LaunchCV AI Authentication Screen & Central Candidate Profile Hub]

Drop screenshot of Auth.jsx login screen (One Text Box / OTP) and central Profile Management Hub

Figure 7.1: LaunchCV AI Authentication Screen & Central Profile Hub

7.2 Remo AI: Interactive Resume Studio & Live Vector Preview

Split-screen editing environment with dynamic text synchronization, font/color selectors, AI refinement sliders, and live A4 PDF page rendering:

[□□ Figure 7.2: Remo AI Resume Studio — Live Editor & PDFKit Dynamic Vector Preview]

Drop screenshot of Remo AI editor showing form controls, AI refine sliders, and dynamic A4 PDF preview

Figure 7.2: Remo AI Resume Studio — Interactive Split Screen Editor & Vector Preview

7.3 Folio AI: Multi-Theme Portfolio Generator & Template Picker

Visual theme gallery allowing users to select Bento Grid, Cyber Terminal, Executive, or Glassmorphism templates and download zero-dependency ZIP packages:

[□□ Figure 7.3: Folio AI Portfolio Studio — Multi-Theme Visual Showcase & ZIP Exporter]

Drop screenshot of Folio AI template selection cards and exported standalone web portfolio preview

Figure 7.3: Folio AI Portfolio Studio — Multi-Theme Visual Showcase & ZIP Exporter

7.4 Mali AI: HTML Email Campaign Studio & Scheduled Queue

Visual rich-text email composer with Google typography picker, theme presets, simulated browser sandbox, and scheduled background queue monitor:

[□□ Figure 7.4: Mali AI & Covo AI — Visual HTML Composer & Scheduled Queue Manager]

Drop screenshot of Mali AI email composer, typography settings, and scheduled campaign queue list

Figure 7.4: Mali AI & Covo AI — Visual HTML Email Composer & Scheduled Queue Dashboard

CHAPTER 8

CONCLUSION, MSBTE CURRICULUM OUTCOMES & FUTURE SCOPE

8.1 Project Conclusion

The development of LaunchCV AI represents a comprehensive, production-grade implementation of full-stack software engineering, generative AI orchestration, and deterministic document compilation. Developed under the MSBTE 5th Semester 'K' Scheme Computer Engineering curriculum for Software Project Implementation (SPI), the project successfully resolves real-world challenges in resume tailoring, ATS optimization, web portfolio generation, and recruiter outreach.

By integrating React 18, Node.js, Express, Supabase PostgreSQL, Groq LPU sub-second inference, and PDFKit mathematical coordinate rendering, LaunchCV AI delivers a unified SaaS workflow that operates reliably across both cloud-connected and offline environments.

8.2 Alignment with MSBTE Course Outcomes (COs) & Program Outcomes (POs)

- CO-1 (Requirements & Analysis): Formulated structured Software Requirement Specifications (SRS) and conducted comparative literature reviews.
- CO-2 (System Design & Modeling): Designed decoupled client-server architectures, Data Flow Diagrams (DFDs), and relational/JSONB database schemas.
- CO-3 (Software Implementation): Mastered modern JavaScript ES6+, React component hierarchies, RESTful Express APIs, and third-party AI integrations.
- CO-4 (Testing & Quality Assurance): Executed comprehensive unit, integration, and security test cases with 100% validation success.

8.3 Future Enhancements & Scope

- Multi-Language Resume Translation: Automated translation of resumes and outreach pitches into major Indian and international languages.
- AI Voice Mock Interviews: Real-time interactive voice simulation conducting behavioral and technical interviews tailored to specific JDs.
- Institutional Placement Portals: Multi-tenant administrative dashboards for polytechnic colleges and universities to monitor student placement progress.

CHAPTER 9

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